

Enhancing the Immunisation Supply Chain with Effective Vaccine Management Analytics

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Outline

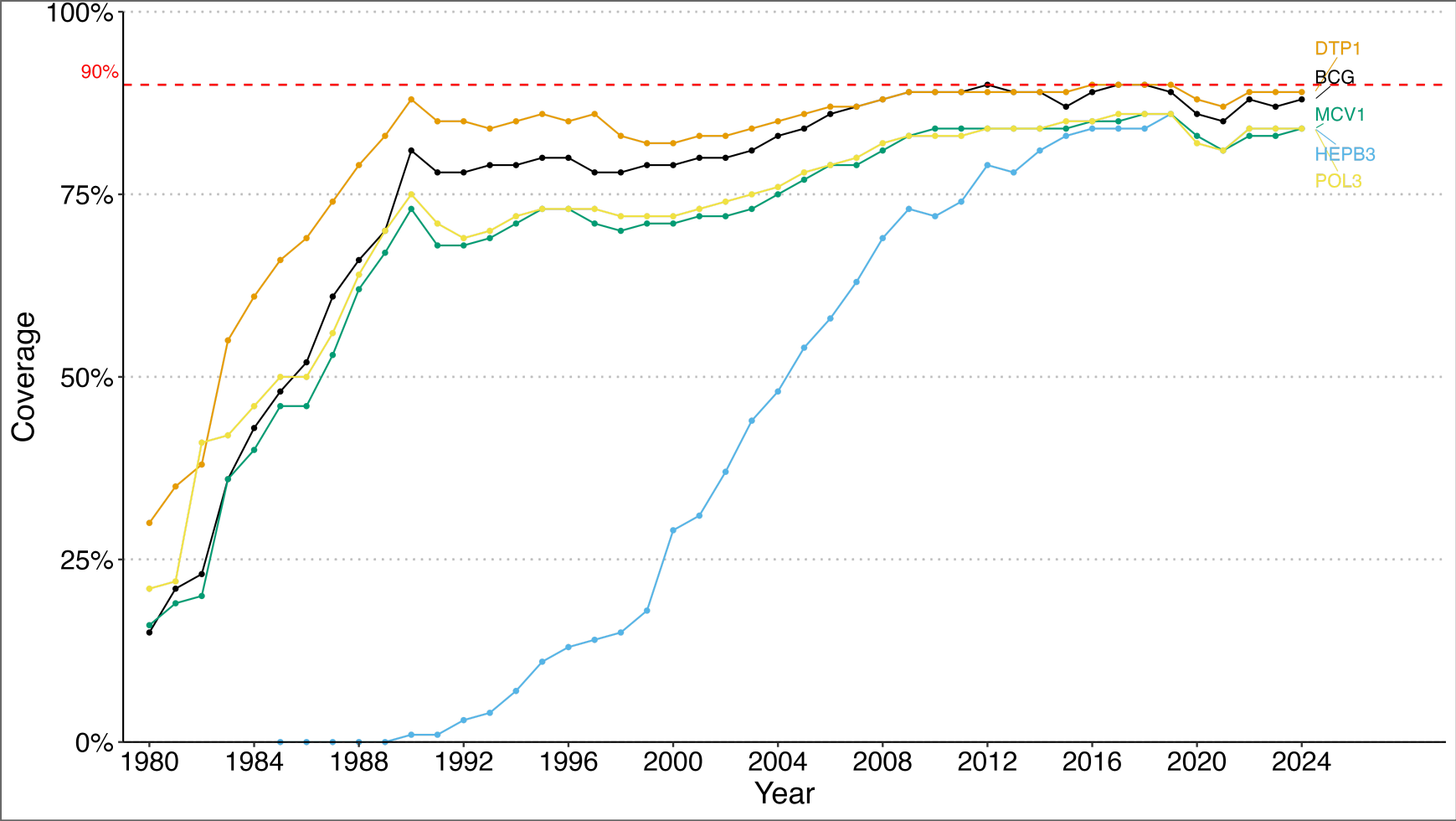
- Immunisations coverage and disparities
- Immunisation Supply Chain (iSC) importance and performance
- Effective Vaccine Management (EVM) initiative
- Analytics for Improving iSC with EVM

Vaccines save
millions of lives
each year by
protecting
children against
serious and
preventable
disease.

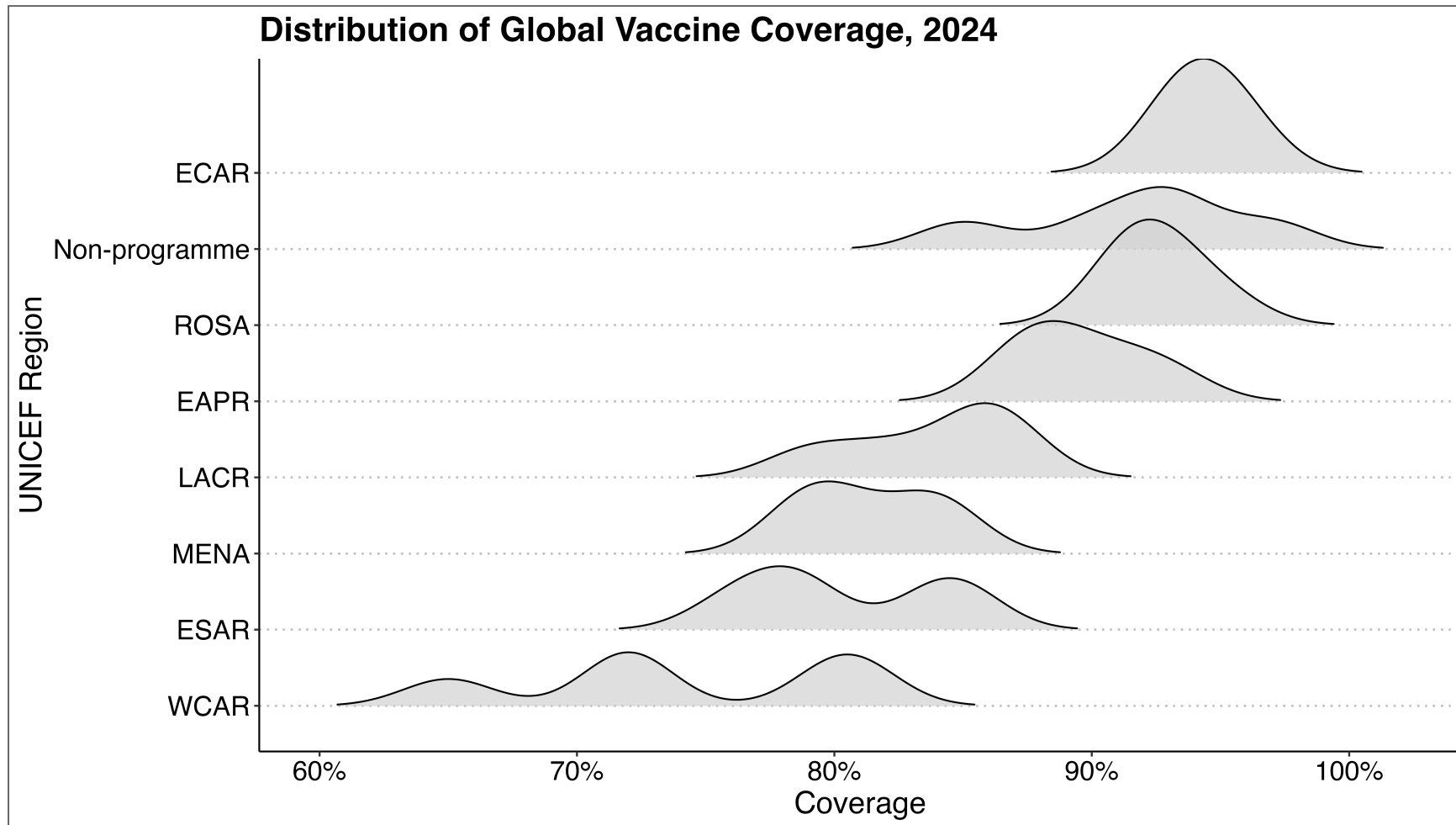
Under-5 deaths worldwide
12.8 million in 1990
4.6 million in 2024

19.9 Million Children Missed Vaccines

4.6 million children died before their 5th birthday



Coverage disparities exist within and between regions



Global support for
increasing immunization

14 of the 17 United Nations
Sustainable Development
Goals demonstrate the social
and economic value of
increasing immunization
coverage.



1. Healthy children + families = **Increased Prosperity**
2. Immunization + nutrition = **Healthier Families**
3. Immunization = **Healthy lives + well-being**
4. Vaccines support cognitive development through better health = **Improved Learning**
5. Immunization = **Empowered Women & Girls**
6. Clean water, sanitation & hygiene (wash) + vaccines = **Less Disease**
7. Efficient supply chain equipment = **Cleaner Environment**
8. Healthy population = **More Productive Workforce**
9. Healthy vaccine market = **Innovation**
10. Better health = **Increased Equality**
11. Protected urban public health = **Healthier Cities**
13. Vaccines = **Mitigation of Climate Change Impact**
16. Strong health systems = **Long-term Stability**
17. Innovative partnership = **Unprecedented Progress**

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The importance of the supply chain

Each successful vaccination depends on a complex immunization supply chain (iSC) to ensure that the correct vaccine is administered at the right **cost, time and place**.

The Gavi Alliance iSC Strategy

The performance of the immunization supply chain is defined by:

- the **availability** of vaccines
 - the **quality** of those vaccines
 - the **efficiency** of the system
-

Source of image: **WHO EVM Document**

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A global initiative to improve immunization supply chain.

A national planning process endorsed and supported by WHO and UNICEF
EVM provides materials and tools needed to monitor and assess vaccine supply chains and help countries improve their supply chain performance.

EVM Framework

The EVM assessment framework defines the way in which vaccine supply chain systems are assessed.

It is organised by:

- **Criteria:** the operational or management functions that health facilities must perform; define what area of the supply chain is being assessed: Vaccine arrivals (E1), temperature management (E2), and waste management (E9).
- **Categories:** determine the necessary inputs, outputs, and performance required to execute the functions defined by the Criteria.
- **Inputs (C1–C6):** What is required to perform the function, such as Infrastructure (C1), Equipment (C2), and Financial Resources (C6).
- **Outputs (O):** The deliverables produced by staff, such as records, charts, or reports.
- **Performance (P):** The outcomes of proper implementation, such as vaccine availability and the absence of heat-damaged vaccines.

EVM Framework

The EVM assessment framework defines the way in which vaccine supply chain systems are assessed.

It is organised by:

- **Criteria:** Define what area of the supply chain is being assessed
- **Categories:** determine the necessary inputs, outputs, and performance required to execute the functions defined by the Criteria.
- **Requirements:** specific minimum standards, or attributes that a well-functioning immunisation supply chain must have; Human Resources (R0136): “iSC [immunization Supply Chain] staff have work mobile phones”
- **Questions:** Provide the data to score Requirements: Interviewing, Verifying, Observing, Calculating.

Scoring in EVM is determined by this assessment framework.

EVM Dataset Overview

- Assessments from 82 countries between 2019 and 2024.
- 4287 health facilities (rows).
- 1108 variables (columns).
- 1286 potential questions.
- 871 requirements, 19 criteria and 8 categories.
- Scores range from 0 to 5.

EVM Framework Guide

R0001: The facility has a functional landline telephone.

E2: Temperature management

M1: Annual needs forecasting

ST: Strategic planning

C1: Infrastructure

C3: Information technology

Data - Scores

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Project objectives

- Exploratory analysis to diagnose system fragility and performance drivers.
- Explain performance and coverage using causal machine learning models.
- Develop a prescriptive optimisation framework that converts analytical findings into actionable guidance for policymakers and stakeholders.

Expected Impact

The output of this project will support policymakers in allocating resources more effectively, and selecting high-impact interventions.

For this talk

- Exploratory analysis to diagnose system fragility and performance drivers.
- Explain performance and coverage using causal machine learning.
- Develop a prescriptive optimisation framework that converts analytical findings into actionable guidance for policymakers and stakeholders.

Dimension Reduction

Empirical Cumulative Distribution Function (ECDF)

Service Points Showing Lowest Availability

Last-Mile Fragility in iSC

Persistent
Underperformance
and High Variability
at Service Points

Any questions or comments? 

References

- › Effective Vaccine Management (EVM) Initiative
- › UNICEF Immunization Data

Image credit: Unsplash

Speaker notes